

Experiment Plan — ODAS + YAMNet Improvement

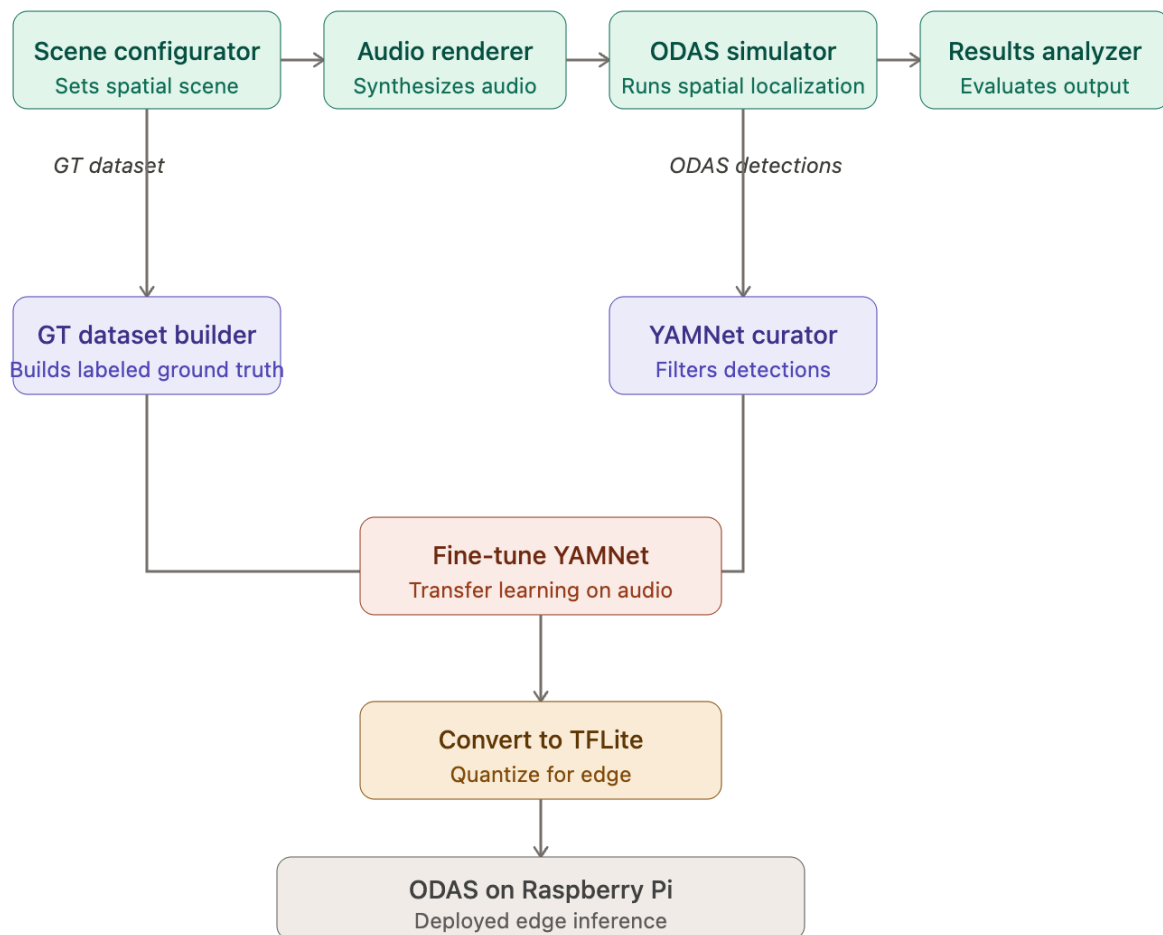
Purpose: Systematic experiments to improve YAMNet classification accuracy and reduce false positives when ODAS is deployed live on a Raspberry Pi.

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Last updated: 2026-05-02

Background & Problem Statement

The full pipeline is:



Two training-data sources exist and differ significantly in distribution:

Source	How produced	Model sees at train time	Model sees at deploy time
GT dataset	Clip extracted directly from rendered audio sidecar (.f32); clean RIR, no beamforming	Pre-beamformed, clean	Post-beamformed, ODAS-filtered
Post-ODAS dataset	Griffin-Lim reconstruction of ODAS .bin spectra; output of the actual beamformer	Post-beamformed, ODAS-filtered	Post-beamformed, ODAS-filtered ✓

Core hypothesis: A model trained on post-ODAS data should transfer better to live deployment because the training distribution matches the inference distribution exactly.

Key problems observed:

1. **False positives (~0.63/s even after SST tuning):** ODAS structural hotspot azimuths (-90° , 0° , -120°) produce spurious tracks during ambient noise. YAMNet has never seen the spectral signature of these ambient-driven ghost tracks during training.
2. **Per-label miss rate varies widely:** `drone_bebop` $\approx 3\%$ miss vs `bear` $\approx 62\%$ miss. Short/quiet events are disproportionately hurt.
3. **Domain gap:** GT-trained model sees pre-beamformed audio; deployed model sees post-beamformed ODAS spectra — different spectral shape.
4. **Real-capture ambient has directional components:** Raw file used as ambient source may contain directional noise that bleeds into the SSL hotspot directions, increasing FPs. Whether this matters systematically has not been tested.

Fixed Protocol (all experiments)

To ensure results are comparable, the following are **fixed** across all experiments in Phase 1:

Parameter	Fixed value
ODAS SST config preset	Balanced (<code>N_prob=6</code> , <code>theta_prob=0.65</code> , <code>Pnew=0.06</code>)
Mic array	ReSpeaker 4-mic square, 64mm
Scene labels	Elephant, Bear, Frog, Lion, <code>drone_bebop</code> , <code>drone_binary</code>
Source distances	10–150m range

Parameter	Fixed value
Scene duration	600s per render
Render room	250×250×20m, absorption=0.7, max_order=3
Training recipe	Phase-1: 20 epochs (head only); Phase-2: 20 epochs (top-4 backbone unfreeze); batch=32
Evaluation	Clip-level test accuracy + event-level precision/recall + FP/min on dedicated hold-out render

Reproducibility: Every scene uses a saved scene JSON. Every training run records the dataset path, preset name, and experiment tag in the run/checkpoint metadata.

Phase 1 — Ambient-Type Ablation

Question: Does the type of ambient background determine how many false positives ODAS generates, and does training on matched ambient improve accuracy?

EXP-A1 — Baseline: No Ambient

Field	Value
Experiment tag	<code>exp_a1_no_ambient</code>
Scene type	Directional sources only, no ambient sources, no capture
Training data	Post-ODAS curator dataset from this scene
Label strategy	GT-only
Expected outcome	Low FP rate (structural hotspots only from sensor noise); provides noise floor baseline
Acceptance criterion	FP/min ≤ 2

Steps:

1. Create scene: Easy mode, 6 labels, 600s, max_radius=150m, **ambient mode = none** (remove all ambient sources)
2. Render → ODAS Simulate (tag: `exp_a1_no_ambient`)
3. Analyze → GT-only label strategy → curate to dataset `odas_a1_no_ambient`
4. Train: GT dataset `gt_a1_no_ambient` only → checkpoint `model_a1_gt`
5. Train: Post-ODAS dataset `odas_a1_no_ambient` only → checkpoint `model_a1_odas`
6. Evaluate both on a **separate** 300s hold-out render (same scene family, different RNG seed)

7. Record: clip acc, event P/R, FP/min

EXP-A2 — Single Synthetic Ambient: Rain

Field	Value
Experiment tag	<code>exp_a2_rain</code>
Scene type	Directional sources + synthetic Rain ambient (volume 0.3–0.5)
Training data	Post-ODAS curator dataset
Label strategy	GT-only
Expected outcome	Rain is spectrally broadband — should create moderate FP rate. Structural hotspots appear more clearly against flat noise.

Steps: Same as EXP-A1 but add **Rain** as synthetic ambient. Use the same fixed SST preset.

EXP-A3 — Single Synthetic Ambient: Wind

Same as EXP-A2 but with **Wind** ambient. Wind is low-frequency heavy — may activate GCC-PHAT at different azimuths than rain.

Experiment tag: `exp_a3_wind`

EXP-A4 — Mixed Synthetic Ambient (Bird + Rain + Wind)

Field	Value
Experiment tag	<code>exp_a4_mixed_synth</code>
Scene type	Directional sources + 3 synthetic ambients (Bird, Rain, Wind), all volume 0.3
Rationale	Closer to natural outdoor environment without directional contamination

EXP-A5 — Real Capture Ambient (Raw File)

Field	Value
Experiment tag	exp_a5_capture_ambient
Scene type	Directional sources + real 4-channel capture ambient
Rationale	Tests the suspected problem: real captures contain directional components that bleed into the SSL hotspot directions
Key comparison	EXP-A4 vs EXP-A5 with equivalent ambient SPL: if FP rate is higher in A5, directional contamination in captures is confirmed

Phase 1 Comparison Matrix

Experiment	Ambient type	Expected FP/min	GT model clip acc	ODAS model clip acc
A1	None	~2	?	?
A2	Rain (synth)	~5	?	?
A3	Wind (synth)	~5	?	?
A4	Mixed synth	~8	?	?
A5	Real capture	~15	?	?

Fill in results as experiments complete.

Phase 2 — Training-Data Strategy Ablation

Question: How does the choice of training data (GT vs post-ODAS vs hybrid) affect deployment accuracy?

Use a **fixed scene**: EXP-A4 (mixed synthetic ambient) as the canonical scene family.

EXP-B1 — GT-Only Model

Field	Value
Experiment tag	exp_b1_gt_only

Field	Value
Training data	GT Dataset Builder clips (pre-beamformed, clean RIR)
Expected weakness	Domain gap — training distribution $\neq$ inference distribution

EXP-B2 — Post-ODAS-Only Model

Field	Value
Experiment tag	<code>exp_b2_odas_only</code>
Training data	Post-ODAS curator dataset (Griffin-Lim from <code>.bin</code> spectral buffers)
Expected strength	Matches deployment distribution; YAMNet sees exactly what it will see live
Expected weakness	May have fewer clips (ODAS misses ~39% of events per Phase 1 tuning)

EXP-B3 — Hybrid: GT Pretrain + ODAS Fine-tune

Field	Value
Experiment tag	<code>exp_b3_hybrid</code>
Training recipe	Warm-start from <code>model_b1_gt</code> checkpoint → fine-tune on post-ODAS dataset
Rationale	GT gives more diverse coverage; ODAS fine-tune corrects the domain gap

EXP-B4 — Post-ODAS + Hard Negatives

Field	Value
Experiment tag	<code>exp_b4_hard_negatives</code>
Training data	Post-ODAS dataset (B2) + ambient-only ODAS peaks labeled <code>background</code>
Ambient-only pipeline	Create scene with zero directional sources , render, simulate, use "Label as background" curator mode in Analyzer

Field	Value
Rationale	Teach the model that ODAS ghost tracks from ambient noise look different from real animal sounds

Ambient-only scene setup:

1. Create scene: 0 directional sources, ambient **Rain+Wind+Bird** at volume 0.5, duration 600s
2. Render → ODAS Simulate (tag: **exp_b4_ambient_only**)
3. In Analyzer → Dataset Curation Settings → enable **"Label all ODAS peaks as background"**
4. Analyze → all detections saved as **background** class to dataset **odas_hard_negatives**
5. Merge **odas_hard_negatives** with **odas_b2_main** → train → checkpoint **model_b4**

EXP-B5 — Multi-Background Pooled Model

Field	Value
Experiment tag	exp_b5_pooled
Training data	Post-ODAS datasets from EXP-A1 through EXP-A5 merged + hard negatives from B4
Rationale	Exposure to diverse ambient conditions during training → robust deployment
Acceptance criterion	Better test acc on unseen ambient condition than any single-ambient model

Phase 2 Comparison Matrix

Model	Training data	Clip acc (synth ambient)	Clip acc (capture ambient)	Event P/R	FP/min
B1 — GT only	GT clips	?	?	?	?
B2 — ODAS only	Post-ODAS	?	?	?	?
B3 — Hybrid	GT pretrain + ODAS FT	?	?	?	?
B4 — +Hard neg	Post-ODAS + ambient BG	?	?	?	?

Model	Training data	Clip acc (synth ambient)	Clip acc (capture ambient)	Event P/R	FP/min
B5 — Pooled	All conditions merged	?	?	?	?

Phase 3 — Confidence Threshold & Deployment Tuning

Once the best model from Phase 2 is identified:

EXP-C1 — Threshold Sweep

Use the Deployment Evaluation tab in the Analyzer to sweep the class-confidence accept threshold (0.3 → 0.95 in steps of 0.05) and record:

- Precision and Recall at each threshold
- FP/min at each threshold
- Choose operating point: max F1 while FP/min $\leq$ 0.1

EXP-C2 — Per-Class Threshold Tuning

Some classes (Elephant, drone_bebop) are easy; others (Bear, Frog) are hard. Set per-class thresholds based on Phase 2 confusion matrices.

EXP-C3 — ODAS Config Preset Comparison

Compare **High-Recall** vs **Balanced** vs **Low-FP** presets (available in the Simulator page) on the B4 model. This separates ODAS config effects from model quality effects.

Preset	N_prob	theta_prob	Pnew	Expected trade-off
High-Recall (dataset collection)	3	0.60	0.15	More events detected, more FPs; use only for building datasets
Balanced (default)	6	0.65	0.06	Current best balance
Low-FP (deployment)	8	0.75	0.03	Fewer FPs, more misses; only use with a model trained on post-ODAS data

Phase 4 — Real-World Validation

After Phase 3 identifies the best model + threshold + ODAS preset:

1. Export best TFLite model from Fine-Tune YAMNet → Deploy tab
2. Deploy to Raspberry Pi with [odaslive](#) using the Low-FP preset config
3. Record 10 minutes of each class from $\geq 30\text{m}$ distance against natural ambient
4. Label detections manually
5. Compute: event precision, event recall, mean azimuth error, FP/min
6. Compare against Phase 3 simulator predictions to validate simulation fidelity

Appendix A — Metrics Definitions

Metric	Definition
Clip accuracy	% of held-out test clips classified with correct label (standard multi-class accuracy)
Event precision	GT events with ≥ 1 matching detection / total detection groups (direction + time aligned)
Event recall	GT events with ≥ 1 matching detection / total GT events
FP/min	Detections during periods with zero active GT sources, normalised by quiet-time duration
Correct-class + correct-direction	Detections where YAMNet class = GT label AND azimuth error $\leq 15^\circ$

Appendix B — App Knobs Reference

Scene Configurator

- **Ambient mode:** Synthetic vs Real Capture — use Synthetic for controlled ambient ablations
- **Source volume:** Per-source gain — use to control SNR
- **Source distance:** Directly controls arrival SNR at array

ODAS Simulator

- **SST preset:** High-Recall / Balanced / Low-FP — must be fixed within an experiment batch
- **Experiment tag:** Propagates to run JSON and downstream datasets/models for traceability

Results Analyzer — Dataset Curation Settings

- **Label strategy:** **GT-only** (Phase 1/2 training), **ODAS event voting** (iterative improvement), **Fine-tuned model** (iterative improvement after first model)
- **Label all ODAS peaks as background:** Enabled only for ambient-only runs (EXP-B4 hard negatives)

Fine-Tune YAMNet — Dataset Tab

- **Max clips per class:** Cap imbalanced classes (e.g., cap Elephant at 200 so rare Bear clips are not swamped)
- **Background injection:** Inject GT background clips when post-ODAS dataset lacks a background class

Appendix C — Known Issues / Constraints

Issue	Impact	Mitigation
Flat array → all sources reported at elevation $\approx 0^\circ$	Cannot separate elevated sources from ground-level	Use only horizontal sources in scenes
GCC-PHAT hotspots at -90° , 0° , -120°	Structural FP floor $\sim 0.6/s$ irreducible by SST tuning alone	Hard-negative training (EXP-B4) + confidence filtering
Griffin-Lim reconstruction artefacts in .bin →WAV	Post-ODAS WAVs sound metallic/choppy	Use .bin directly for training; WAV is for human verification only
Short events (<2s) disproportionately missed at N_prob=6	Bear, Frog high miss rate	Use High-Recall preset for dataset collection only
Source files repeat across scenes	May cause clip-level train/test leakage	max_clips_per_class cap + source-file grouped split in finetuner